

Hoang-An LE | Doctoral candidate

Info

Born: June 16, 1990

Nationality: Vietnamese

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Research interest: Multi-modality Computer Vision with Deep Learning, including intrinsic image decomposition (photometry), depth and surface normal (geometry), optical flow (motion), and object labeling (semantics).

Education

Doctor of Philosophy in Computer Science

University of Amsterdam, the Netherlands

funded by TrimBot2020 project, EU Horizon program

2016–present

- **Advisor** Prof. Theo Gevers and A/Prof. Thomas Mensink
- *Provisional graduation* Fall 2020
- *Tentative dissertation* Multi-modality Computer Vision

Master of Science in Engineering | Diplôme d'Ingénieur

Télécom ParisTech, France

international program—all classes in English

2012–2015

- 2012–2013 | *John von Neumann institute*, Vietnam National University, Vietnam
- 2013–2015 | *EURECOM institute, Campus SophiaTech*, Télécom ParisTech, France
 - **GPA** 4.00/4.00 | **Eiffel scholarship** by Campus France
 - **Semester project** “Person-based Video Hyperlinking” **Advisor** A/Prof. Benoit Huet
Recognizing and linking people in a video dataset based on LBP facial descriptors.
 - **Graduation project** “Multi-Camera Matching and Point Clouds Registration for 3D Immersive Visual Systems”
Advisor Prof. Minh Do, UIUC, USA.

Bachelor of Science

Ho Chi Minh University of Science, Vietnam

advanced program in computer science—all classes in English

2008–2012

- **GPA** 3.96/4.00 | **Rank** 1/22 (in class), 1/600 (among intakes)
- **Graduation project** “Natural User Interface for Smart Environment” | **Mark** 10.0/10.0
Advisor A/Prof. Minh-Triet Tran
Designing and implementing a smart multimodal environment supporting natural body gestures, voices and handheld devices using Microsoft Kinect. The system is published in proceeding of HCII’13.

Notable Courses

Marco Loog, David Tax

TU Delft

Advanced Pattern Recognition, the Netherlands

2017

Arnold Smeulders, Cees Snoek, Efstratios Gavves

University of Amsterdam

Computer Vision by Learning, the Netherlands

2017

Prof. Bernard Merialdo

EURECOM

Intelligent Systems, France

2014

Prof. Bernard Merialdo

EURECOM

Multimedia Indexing and Retrieval, France

2013

Prof. Bao T. Ho

JAIST

Data Mining and Information, Japan

Experiences

Project.....

TrimBot2020 Cutting Hedge Research

University of Amsterdam

funded by European Union Horizon 2020 program, the Netherlands

Apr 2016–Dec 2020

Supervisor Prof. Theo Gevers

The TrimBot2020 project develops a vision-controlled robot for autonomous navigation and trimming of gardens, which works in uncontrolled outdoor conditions without the need for active illumination or sensors. The robot navigates over varying terrain and approaches rose bushes and boxwood topiary, to trim them to an ideal shape. We are responsible for the vision-based 3D data analysis and scene understanding work-package. The project consortium partners include University of Edinburgh, Wageningen University and Research, ETH Zurich, Albert-Ludwigs-University Freiburg, Rijksuniversiteit Groningen, and Robert Bosch company.

Internship.....

Multi-Camera Matching and Point clouds Registration

CSL@UIUC and Personify Inc.

Obiwan project: 3D Immersive Visual System, Illinois, USA

Mar 2015–Sep 2015

Supervisor Prof. Minh Do and Prof. Sanjay Patel

A joint project between Coordinate Science Lab (UIUC) and Personify Inc. to study and implement an end-to-end system that provides full 3D immersive experience using Intel's newest released RealSense camera R200 and Oculus Rift DK2.

Search and Hyperlinking task at MediaEval 2014

EURECOM institute

LinkedTV project: Multimedia and Video retrieval, France

Jun 2014–Sep 2014

Supervisor A/Prof. Benoit Huet

The search and hyperlinking task requires participants to retrieve relevant video segments, given texts or videos as search queries. The interns participated as members of the LinkedTV team to propose solutions for the tasks. The internship was concluded by a working note published in the workshop proceeding and a paper in SLAM'15.

Teaching Assistant.....

University of Amsterdam

Computer Vision 1, Amsterdam, the Netherlands

2017–present

Lecturer Prof. Theo Gevers

University of Amsterdam

Computer Vision 2, Amsterdam, the Netherlands

2016–2017

Lecturer Prof. Theo Gevers

Honors and awards

Being interviewed for the TrimBot2020 project's *Human Face of Robotics* 2020
<https://tinyurl.com/tb2020-hale>

Oral presentation at BMVC, *Three for One and One for Three: Flow, Segmentation, and Surface Normal* 2018

Eiffel scholarship by Campus France 2013–2015

Award in Nokia Tap The Apps contest 2011

Graduated at rank 1/22 of class and 1/600 among intake 2012

Academic Services

Reviewer for CVPR 2020

Reviewer for BMVC 2019

Volunteer for ECCV 2016

Publications

Journals.....

- [J4] **HA. Le**, T. Nimbhorkar, T. Mensink, AS. Baslamisli, S. Karaoglu, T. Gevers, “Unsupervised Generation of Optical Flow Datasets”, *under submission to IEEE Transactions of Image Processing*, 2020.
- [J3] AS. Baslamisli, P. Das, **HA. Le**, S. Karaoglu, T. Gevers, “ShadingNet: Image Intrinsic by Fine-Grained Shading Decomposition”, *under submission to IEEE Transactions of Image Processing*, 2020.
- [J2] J. Han, S. Karaoglu, **HA. Le**, T. Gevers, “Improving Face Detection Performance with 3D-Rendered Synthetic Data” *under submission to Pattern Recognition Letters, Elsevier*, 2020.
- [J1] **HA. Le**, MT. Doan, MT. Tran, “Webcam-based Laser Dot Detection Technique in Computer Remote Control”, in *Proceedings of the 3th International Conference on Theories and Applications of Computer Science: Special issue of The Journal of Science and Technology (ICTACS)*, 2010.

International Conferences.....

- [C8] **HA. Le**, P. Das, T. Mensink, T. Gevers, “Geometry-based Novel View Synthesis from Single Image”, *under submission to European Conference in Computer Vision (ECCV)*, 2020.
- [C7] **HA. Le**, AS. Baslamisli, T. Mensink, T. Gevers, “Three for one and one for three: Flow, Segmentation, and Surface Normals”, in *British Machine Vision Conference (BMVC)*, 2018 **Oral** (acceptance rate 4.3%).
- [C6] AS. Baslamisli, TT. Groenestege, P. Das, **HA. Le**, S. Karaoglu, T. Gevers, “Joint Learning of Intrinsic Images and Semantic Segmentation”, in *European Conference in Computer Vision (ECCV)*, 2018.
- [C5] AS. Baslamisli, **HA. Le**, T. Gevers, “CNN based Learning using Reflection and Retinex Models for Intrinsic Image Decomposition”, in *Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [C4] M. Eskevich, QM. Bui, **HA. Le**, B. Huet, “Exploring Video Hyperlinking in Broadcast Media”, in *Proceedings of the Third Edition Workshop on Speech, Language and Audio in Multimedia (SLAM)*, 2015.
- [C3] **HA. Le**, KNC. Mac, TA. Pham, VT. Nguyen, MT. Tran, “Multimodal Smart Interactive Presentation System”, in *Proceedings of the 15th International Conference on Human-Computer Interaction (HCII)*, 2013.
- [C2] **HA. Le**, KNC. Mac, TA. Pham, MT. Tran, “Realtime Pointing Gesture Recognition and Applications in Multi-user Interaction”, in *Proceedings of the 5th Asian Conference on Intelligent Information and Database Systems (ACIIDS)*, 2013.
- [C1] **HA. Le**, KNC. Mac, TA. Pham, VT. Nguyen, MT. Tran, AD Duong, “SIM - Smart Interactive Map with Pointing Gestures”, in *Proceedings of the 4th International Conference on Intelligent Human-Machine Systems and Cybernetics, IEEE Computer Society's Conference (IHMSC)*, 2012.

Workshops.....

- [W3] R. Tylecek, T. Sattler, **HA. Le**, T. Brox, M. Pollefeys, RB. Fisher, T. Gevers, “The Second Workshop on 3D Reconstruction Meets Semantics: Challenge Results Discussion”, in *European Conference in Computer Vision Workshop (ECCVw)*.
- [W2] I. Pratikakis, MA. Savelonas, F. Arnaoutoglou, G. Ioannakis, A. Koutsoudis, T. Theoharis, MT. Tran, VT. Nguyen, VK. Pham, HD. Nguyen, **HA. Le**, BH. Tran, QH. To, MB. Truong, TV. Phan, MD. Nguyen, TA. Than, KNC. Mac, MN. Do, AD. Duong, T. Furuya, R. Ohbuchi, M. Aono, S. Tashiro, D. Pickup, X. Sun, PL. Rosin, RR. Martin, “Partial Shape Queries for 3D Object Retrieval”, in *Eurographics 2016 Workshop on 3D Object Retrieval*, 2016.
- [W1] **HA. Le**, QM. Bui, B. Huet, B. Cervenкова, J. Bouchner, E. Apostolidis, F. Markatopoulou, A. Pournaras, V. Mezaris, D. Stein, S. Eickeler, M. Stadtschnitzer “LinkedTV at MediaEval 2014 Search and Hyperlinking Task”, in *Working Notes Proceedings of the MediaEval 2014 Workshop*, 2014.